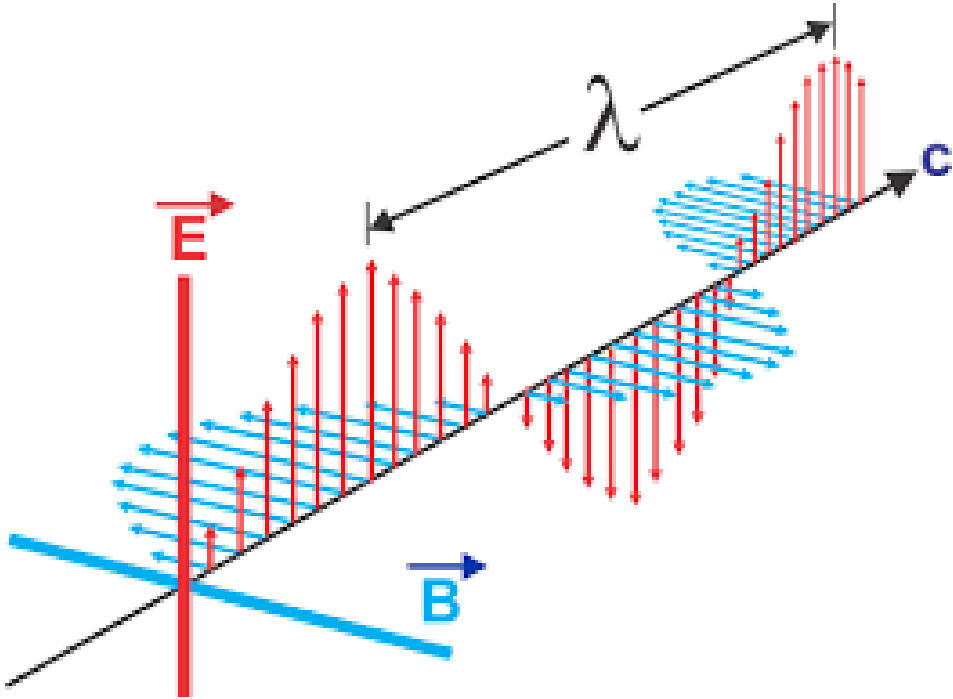


Electromagnetic Wave Propagation & Wave Nature of Light

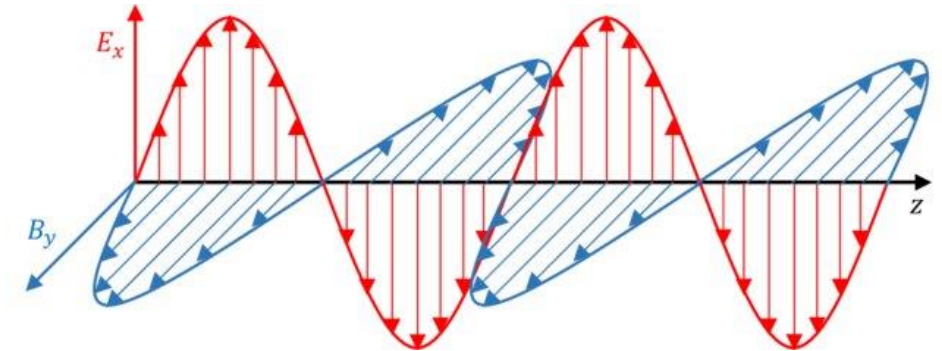
Lecture-1



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Electromagnetic **W**ave (or **L**ight) **P**ropagation in **D**ifferent **M**ediums

1.In vacuum

2.In matter

3.In conductor

4.Waveguide

5.EM wave interactions in different mediums

Electromagnetic waves in vacuum

Maxwell's equations (wave equation for $\vec{E}$ and $\vec{B}$)

$\vec{\nabla} \cdot \vec{D} = \rho_f$		$\vec{D} = \epsilon \vec{E}$
$\vec{\nabla} \cdot \vec{B} = 0$		$\vec{B} = \mu \vec{H}$
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	 (A)	
$\vec{\nabla} \times \vec{B} = \mu \left(\vec{J}_f + \frac{\partial \vec{D}}{\partial t} \right)$		

where $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ and $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

About Maxwell's Equations

“Simple enough to imprint on a T-shirt, and yet reach enough to provide new insights throughout a lifetime of study”

— J. R. Whinnery

“The Teaching of Electromagnetics,” *IEEE Trans. Education*, Vol. 33, pp. 3-7 (1990)

For free space,

The charge density $\rho_f = 0$

And the current density $\vec{J}_f = 0$

Also, $\epsilon_r = 1$ and $\mu_r = 1$

Hence, $\epsilon = \epsilon_0$ and $\mu = \mu_0$

In view of above conditions, the Maxwell's equations;

$\vec{\nabla} \cdot \vec{E} = 0$	 (a)	
$\vec{\nabla} \cdot \vec{B} = 0$	(b)	
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	 (c)	 (1)
$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$	 (d)	

Taking curl of eq 1 (c), we get

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\vec{\nabla} \times \frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

$$(\vec{\nabla} \cdot \vec{E})\vec{\nabla} - (\vec{\nabla} \cdot \vec{\nabla})\vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

But as $\vec{\nabla} \cdot \vec{E} = 0$, We get

$$\nabla^2 \vec{E} = \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

Now using equation 1 (d), we have

$$\nabla^2 \vec{E} = \frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \dots \dots \dots (2)$$

Similarly, we will have

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \dots \dots \dots (3)$$

Equation 2 and 3 represents the relation between the space and time variation of $\vec{E}$ and $\vec{B}$ are called the three-dimensional wave equations for $\vec{E}$ and $\vec{B}$

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{B}) &= \nabla(\cancel{\nabla \cdot \mathbf{B}}) - \nabla^2 \mathbf{B} = \nabla \times \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} \\ \Rightarrow \nabla^2 \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} \end{aligned}$$

Now standard wave equation is

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \dots \dots \dots (4)$$

Comparing eqn 2/3 with 4, we find that $\mu_0\epsilon_0$ has the same magnitude as $\frac{1}{v^2}$
i.e. we have

$$\frac{1}{v^2} = \mu_0\epsilon_0$$

$$v = \frac{1}{\sqrt{\mu_0\epsilon_0}} \dots \dots \dots (5)$$

$$v = \frac{1}{\sqrt{\mu_0\epsilon_0}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}} \frac{m}{s}$$

$$= 0.02999 \times 10^{10} m/s$$

$$= 2.999 \times 10^8 m/s$$

$$\cong 3 \times \frac{10^8 m}{s} = c$$

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \dots \dots \dots (6)$$

This is the velocity with which electromagnetic waves travels in free space.

$$\mu_0 = 4\pi \times 10^{-7} \text{Weber/A.m}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2/\text{N.m}^2$$

⇒ in free space e.m. waves travel with speed of light.

**Notice the crucial role played by Maxwell's contribution to Ampere's law ($\mu_0\epsilon_0\partial\mathbf{E}/\partial t$);
without it, the wave equation would not emerge,
and there would be no electromagnetic theory of light.**

Electromagnetic wave in matter

Maxwell's equation in matter

$$\left. \begin{aligned} \vec{\nabla} \cdot \vec{D} &= \rho_f \\ \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{H} &= \vec{J}_f + \frac{\partial \vec{D}}{\partial t} \end{aligned} \right\} \dots\dots\dots (1)$$

Consider electromagnetic wave inside linear matter, where there are no free charges or free currents

(Linear medium is an insulator or a non-conductor)

$$\left. \begin{aligned} \vec{J}_f &= 0 \\ \sigma_f &= 0 \end{aligned} \right\} \dots\dots\dots (2)$$

For this case, equation (1) becomes

$$\left. \begin{aligned} \vec{\nabla} \cdot \vec{D} &= 0 \\ \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{H} &= \frac{\partial \vec{D}}{\partial t} \end{aligned} \right\} \dots\dots\dots (3)$$

Now, for linear, homogeneous and isotropic medium,

$$\vec{D} = \epsilon \vec{E} \quad \text{and} \quad \vec{H} = \frac{\vec{B}}{\mu} \quad \dots\dots\dots (4)$$

Therefore, Maxwell's equation for E and B fields inside a linear, homogeneous and isotropic non-conducting medium become:

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu\epsilon \frac{\partial \vec{E}}{\partial t} \quad \dots\dots\dots (5)$$

These relations are almost identical to those for electromagnetic waves in free space

- Linear: The electric and magnetic responses of the material are directly proportional to the electric and magnetic fields applied to it.
- Homogeneous: The material has uniform properties throughout its volume.
- Isotropic: The material has similar properties in any direction of travel for waves through that medium

$$\text{In vacuum, } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

$$\text{But in linear medium, } v = \frac{1}{\sqrt{\mu\epsilon}}$$

Which differ from the free space case only with the term $\mu\epsilon$ instead of $\mu_0\epsilon_0$.
The velocity of the electromagnetic wave in the material medium;

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

From optics, the refractive index

$$n = \frac{\text{speed of light}}{\text{speed in the material medium}} = \frac{c}{v} = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} = \sqrt{\frac{\mu_r \mu_0 \epsilon_r \epsilon_0}{\mu_0 \epsilon_0}}$$

$$n = \sqrt{\mu_r \epsilon_r}$$

The solution of the wave equation;

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

$$\text{and } \mathbf{B}(\mathbf{r},t) = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

Where $\mathbf{k}$ is the propagation vector and $\mathbf{r}$ is the position vector.

Plane wave propagation in free space (E , B and n relationship)

The electric and magnetic field components of the plane EM wave are given by

$$\nabla^2 E = \mu_0 \varepsilon_0 \partial^2 E / \partial t^2 \dots\dots\dots (1)$$

$$\text{and } \nabla^2 B = \mu_0 \varepsilon_0 \partial^2 B / \partial t^2 \dots\dots\dots (2)$$

$$\text{Or } \nabla^2 E - \frac{1}{v^2} \partial^2 E / \partial t^2 = 0 \dots\dots\dots (3)$$

$$\nabla^2 B - \frac{1}{v^2} \partial^2 B / \partial t^2 = 0 \dots\dots\dots (4)$$

Since E and B are function of space and time, their solution can also be assumed as

$$E(r, t) = E(r) e^{-i\omega t} \dots\dots\dots (5)$$

$$B(r, t) = B(r) e^{-i\omega t} \dots\dots\dots (6)$$

Substituting these values in equation (3) and (4), we will get

$$e^{-i\omega t} \nabla^2 \vec{E}(r) + \frac{\omega^2}{v^2} \vec{E}(r) e^{-i\omega t} = 0$$

$$\text{or } \nabla^2 \vec{E}(r) + \frac{\omega^2}{v^2} \vec{E}(r) = 0 \quad \dots\dots (7)$$

$$\text{and } \nabla^2 \vec{B}(r) + \frac{\omega^2}{v^2} \vec{B}(r) = 0 \quad \dots\dots\dots (8)$$

Since ω is angular frequency

$$\frac{\omega}{v} = \frac{2\pi\nu}{\lambda} = \frac{2\pi}{\lambda} = k$$

Where k is the propagation vector

$$\nabla^2 \vec{E}(r) + k^2 \vec{E}(r) = 0 \quad \dots\dots\dots (9)$$

$$\nabla^2 \vec{B}(r) + k^2 \vec{B}(r) = 0 \quad \dots\dots\dots (10)$$

These equations are of type

$$\nabla^2 \varphi(r) + k^2 \varphi(r) = 0 \dots\dots\dots(11)$$

known as Helmholtz equation and $\varphi(r)$ represent one of the six field components .

The electric and magnetic vector obey identical wave equation Viz - Helmholtz equation.

The solution of equation (11) can be written as

$$\varphi(x,y,z) = X(x) Y(y) Z(z)$$

Substituting these values in equation (11), we will get

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) XYZ + k^2 XYZ = 0$$

$$\left(YZ \frac{\partial^2 X}{\partial x^2} + XZ \frac{\partial^2 Y}{\partial y^2} + XY \frac{\partial^2 Z}{\partial z^2} \right) + k^2 XYZ = 0$$

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -k^2 \dots\dots\dots(12)$$

Each term of the left is a constant from the variable separation such that

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -k_x^2$$

$$\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = -k_y^2$$

$$\frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -k_z^2$$

Solution of these equations are

$$X = X_0 e^{\pm i k_x x}$$

$$Y = Y_0 e^{\pm i k_y y}$$

$$Z = Z_0 e^{\pm i k_z z}$$

$$\text{So that } \varphi = X_0 Y_0 Z_0 e^{\pm i(k_x x + k_y y + k_z z)}$$

Hence solution in terms of E field will be

$$E_x = E_{0x} e^{\pm i \vec{k} \cdot \vec{r}}, \quad E_y = E_{0y} e^{\pm i \vec{k} \cdot \vec{r}}, \quad E_z = E_{0z} e^{\pm i \vec{k} \cdot \vec{r}} \quad \text{.. (13)}$$

Same equation for B_x, B_y, B_z

Thus in vector form

$$\vec{E}(\vec{r}) = \vec{E}_0 e^{\pm i\vec{k} \cdot \vec{r}}$$

$$\vec{B}(\vec{r}) = \vec{B}_0 e^{\pm i\vec{k} \cdot \vec{r}}$$

Hence

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{-i(\omega t \mp \vec{k} \cdot \vec{r})}$$

$$\text{and } \vec{B}(\vec{r}, t) = \vec{B}_0 e^{-i(\omega t \mp \vec{k} \cdot \vec{r})}$$

$$\text{Since } \vec{k} = \frac{2\pi}{\lambda} = \frac{2\pi\nu}{v} = \frac{\omega}{v} \hat{n}$$

$$\text{Hence } \vec{E}(\vec{r}, t) = \vec{E}_0 e^{-i\omega(t \mp \frac{\hat{n} \cdot \vec{r}}{v})} \dots\dots\dots(14)$$

$$\text{and } \vec{B}(\vec{r}, t) = \vec{B}_0 e^{-i\omega(t \mp \frac{\hat{n} \cdot \vec{r}}{v})}$$

Where $\hat{n}$ is the unit vector in the propagation direction .

We see that

$$E_x (r,t) = E_{0x} f \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right)$$

$$E_y (r,t) = E_{0y} f \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \dots \dots \dots (15)$$

$$\text{And } E_z (r,t) = E_{0z} f \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right)$$

$$\text{Now } \frac{\partial E_x}{\partial x} = E_{0x} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \left(\frac{-n_x}{v} \right)$$

Where n_x is the x - components of $\vec{n}$

$$\frac{\partial E_x}{\partial x} = \left(\frac{-n_x}{v} \right) E_{0x} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right)$$

Since $\vec{\nabla} \cdot \vec{E} = 0$ in free space

$$-\frac{1}{v} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \{ E_{0x} n_x + E_{0y} n_y + E_{0z} n_z \} = 0$$

$$-\frac{\vec{E}_0 \cdot \hat{n}}{v} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) = 0$$

$$\text{Hence } \vec{E}_0 \cdot \hat{n} = 0$$

It means that $\widetilde{E}_0$ is normal to the direction of propagation .

In the same way,

$$\widetilde{B}_0 \cdot \hat{n} = 0 \dots\dots\dots(17)$$

Thus B_0 is normal to the direction of propagation.

Let us also find relation between $\widetilde{E}_0$ and $\widetilde{B}_0$

$$\text{Since } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$(\vec{\nabla} \times \vec{E})_x = -\frac{\partial B_x}{\partial t}$$

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t} = -\frac{\partial}{\partial t} \left\{ B_{0x} f \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \right\} \dots\dots\dots(18)$$

$$\frac{\partial E_z}{\partial y} = \frac{\partial}{\partial y} \left\{ E_{0z} f \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \right\}$$

$$= -\frac{n_y}{v} \left\{ E_{0z} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \right\}$$

$$\frac{\partial E_y}{\partial z} = -\frac{n_z}{v} \left\{ E_{0y} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \right\}$$

Hence

$$(\vec{\nabla} \times \vec{E})_x = \left(\left(\frac{-n_y}{v} \right) E_{0z} + \left(\frac{n_z}{v} \right) E_{0y} \right) \left\{ f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \right\}$$

$$= \left\{ -B_{0x} f' \left(t - \frac{\vec{r} \cdot \hat{n}}{v} \right) \right\}$$

$$\text{Or } \frac{n_y}{v} E_{0z} - \frac{n_z E_{0y}}{v} = B_{0x}$$

$$\text{Or } n_y E_{0z} - n_z E_{0y} = v B_{0x}$$

$$\text{Similarly, } n_z E_{0x} - n_x E_{0z} = v B_{0y}$$

$$n_x E_{0y} - n_z E_{0y} = v B_{0z}$$

Thus combining three eqs. we will get

$$\hat{n} \times \vec{E}_0 = v \vec{B}_0$$

Hence $\vec{B}_0$ is right angle to both $\vec{E}_0$ and $\hat{n}$

Thus $\vec{E}$ and $\vec{B}$ are mutually perpendicular and transverse to the direction of propagation.

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\vec{B}(\vec{r}, t) = \vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

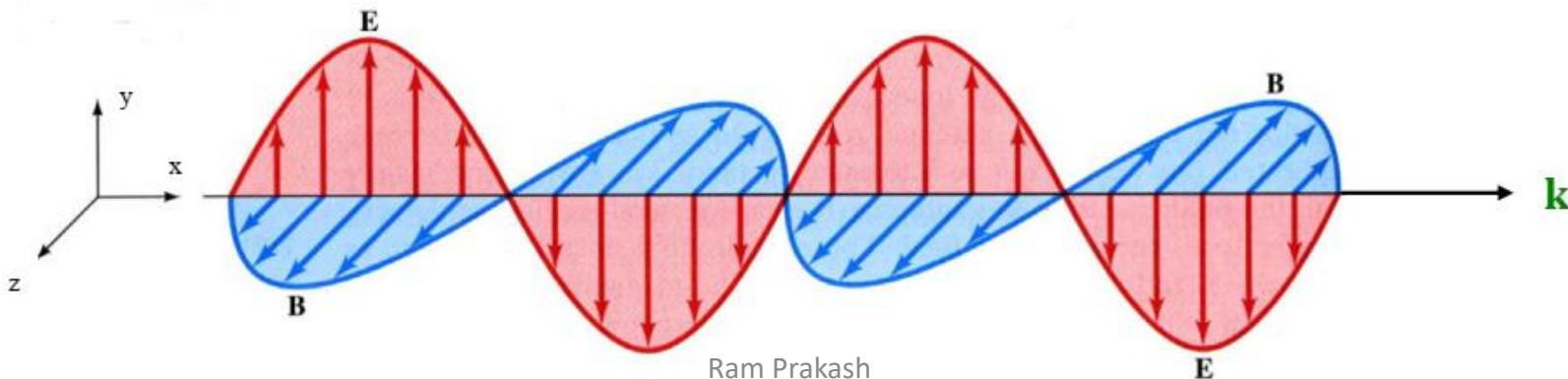
$$\frac{\partial}{\partial t} = -i\omega$$

$$\text{and } \frac{\partial}{\partial x} = i k_x \text{ or } \vec{\nabla} = i \vec{k} \text{ (in case of 3-D)}$$

Algebraic form of Maxwell's Equations in free space

- $i \vec{k} \cdot \vec{E} = 0$ (i.e. $\vec{E}$ is perpendicular to $\vec{k}$)
- $i \vec{k} \cdot \vec{B} = 0$ (i.e. $\vec{B}$ is perpendicular to $\vec{k}$)
- $i \vec{k} \times \vec{E} = i\omega \vec{B}$
- $i \vec{k} \times \vec{B} = -i\epsilon_0\mu_0 \omega \vec{E}$

E and B are mutually perpendicular to each other,
E and B are perpendicular to the direction of propagation of wave.



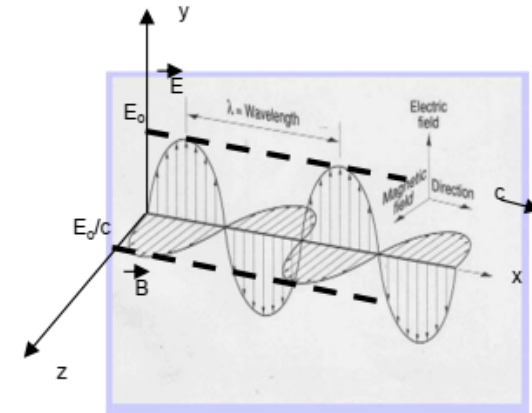
What is the relation between E and B ?

Or can we show that E and B are in same phase at any time in space.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \Rightarrow \mathbf{k} \times \mathbf{E} = \omega \mathbf{B}$$

$$\text{when } \left. \begin{array}{l} E \rightarrow E_y \\ B \rightarrow B_z \\ k \rightarrow x \end{array} \right\} \Rightarrow \left. \begin{array}{l} k E_y = \omega B_z \\ E_y = \frac{\omega}{k} B_z \end{array} \right\} \Rightarrow \left. \begin{array}{l} E_y = c B_z \\ B_z = \frac{E_y}{c} \end{array} \right\}$$

$$\text{But } \left. \begin{array}{l} E_y = E_o e^{i(kx - \omega t)} \\ B_z = B_o e^{i(kx - \omega t)} \end{array} \right\} \Rightarrow \boxed{B_o = \frac{E_o}{c}}$$



Since k/ω is a real number, the electric and magnetic vectors are in phase; thus if at any instant, E is zero then B is also zero, when E attains its maximum value, B also attains its maximum value, etc.

Both E_y and B_z are in same phase.

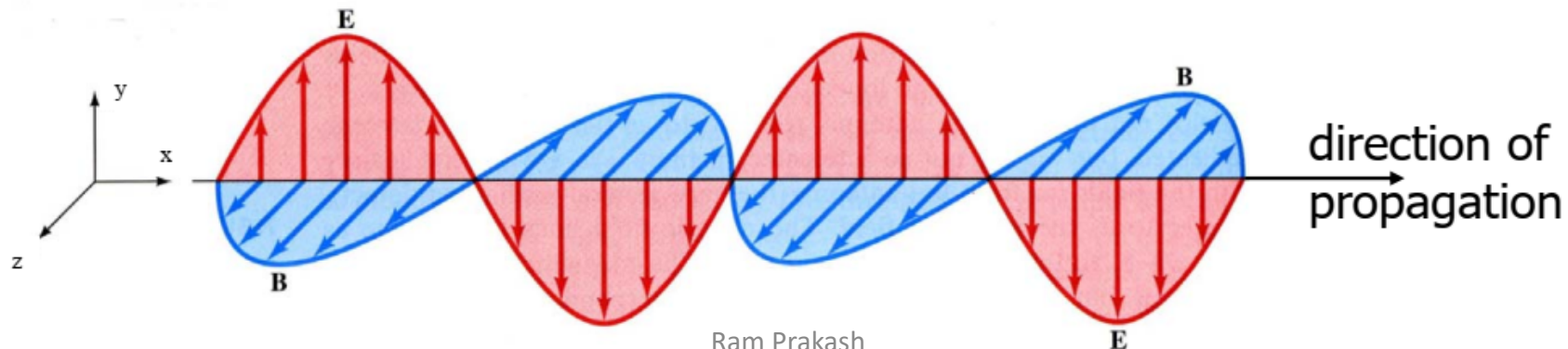
Summary of Important Properties of Electromagnetic Waves

- The **solutions** (plane wave) of Maxwell's equations are wave-like with both E and B satisfying a wave equation.

$$E_y = E_0 \sin(kx - \omega t)$$

$$B_z = B_0 \sin(kx - \omega t)$$

- Electromagnetic waves travel through empty space with the **speed of light** $c = 1/(\mu_0 \epsilon_0)^{1/2}$.
- The plane wave as represented by above is said to be **linearly polarized** because the electric vector is always along y-axis and, similarly, the magnetic vector is always along z-axis.

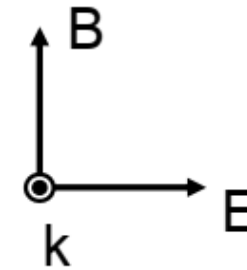


→ The components of the electric and magnetic fields of plane EM waves are perpendicular to each other and perpendicular to the direction of wave propagation. The latter property says that **EM waves are transverse waves.**

→ The magnitudes of E and B in empty space are related by $E/B = c$.

$$\frac{E_0}{B_0} = \frac{E}{B} = \frac{\omega}{k} = c$$

→ Both E and B are at right angles to the direction of propagation. Thus the waves are transverse.



→ The electric and magnetic waves are interdependent; neither can exist without the other. Physically, an electric field varying in time produces a magnetic field varying in space and time; this changing magnetic field produces an electric field varying in space and time and so on.

This mutual generation of electric and magnetic fields result in the propagation of the EM waves.

Thanks